

TIM 172A, MOT-I, Lecture #17 (11/25/25)

Agenda

1. HW #8, Problems 1 & 2
 2. Financial Modeling for new product development (NPD)
 - Discounted Cash Flows (DCF's)
 - Present Value (PV) of a future cash flow
 - Net Present Value (NPV) of expected profit
 - modeling of all relevant cash flows for NPD.
 - NPV basics : derivation of the key formulas
 3. Please work on HW#8, due on the last day of class (L#19)
 4. Final Exam will be handed out on L#19
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1. HW # 8

Prob 1 :

- Carefully study The 'Apple / Intel Product Platform / Line strategy handout

Intel was discussed in L # 16 within the context of the IPMS.

Prob 2 : Create a Prod Platform / Line strategy for the robot designed in HW # 6

- Apply the 5-step IPMS process developed in L # 16 (previous lecture)
- Emulate Intel

2. Financial Modeling

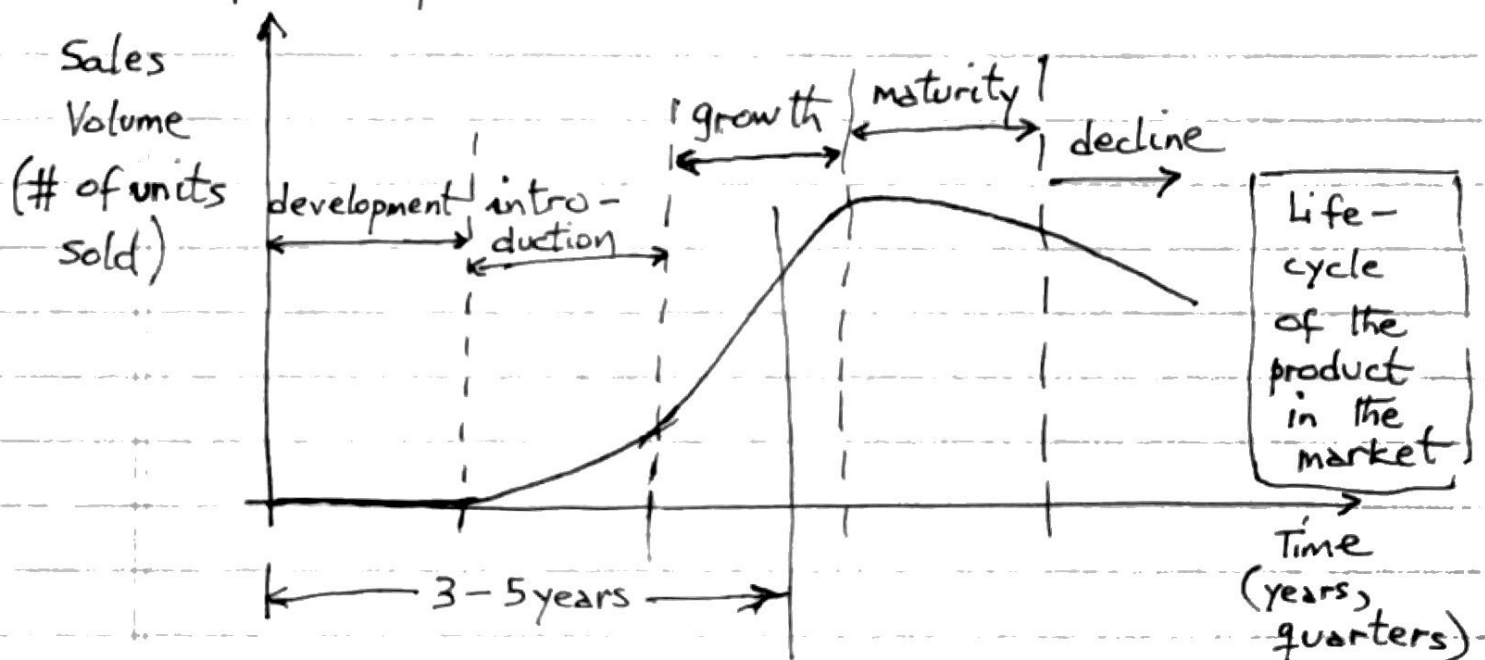
So far

- designed the product
- proto-typed the product
- created the product platform / product lines

But, will the product be successful, i.e. be profitable?

Statement of the problem.

What is the expected profit for "x" years ($x=3,4,5,\dots$) of the life-cycle of the product, starting with the actual development of the product?



For Intel, the life-cycle of a typical microprocessor is 7-8 years [see the "I/A...." handou

References

- chapter on "Product Development Economics" in the PD&D text by U&E
- Product Development Financial modeling tutorial (aka "Excel Tutorial") on Canvas

Process (or Plan) & Execution (or Implementation)

Step 1 : Create a base-case (or nominal) financial model of all relevant cash-flows:

cash "outflows" {

- Development (conceptual design, prototyping, ...)
- Manufacture (or Production)
- Ramp-up cost (to prepare for production: equipment, tooling, ...)
- Marketing & sales costs

cash "inflow" → - Sales Revenue

Step 1(a) : Estimate the timing & magnitude of all the above cash-flows

Step 1(b) : For each quarter, determine the profit or net cash flow.

Example: for YR 3, Q 1 $\equiv$ Quarter 9 ($= 2 \times 4 + 1$)

Marketing = $-\$250,000$

Production = $-\$2,000,000$

Sales Revenue = $+\$4,000,000$

Profit or Net Cash Flow = $\$1,750,000 \equiv \$1,750K$
($\$1K = 1 \text{ Kilodollar} = \1000)

Step 1(c): Compute the present value (PV)
of the profit (or net cash flow) for
each quarter

$d \triangleq$ annual discount factor (%)

Typically, $10\% < d < 15\%$

Quarterly discount factor $d' = \frac{d}{4} (\%)$

EQN.

(1) ←

Present value
of a future
cash flow
(quarterly basis)

$$P_1 = \frac{F_i}{(1 + d')^{i-1}} = \frac{F_i}{\left(1 + \frac{d}{4}\right)^{i-1}},$$

where $F_i \triangleq$ future cash flow for Qtr i ,
($i = 1, 2, 3, \dots - n$)

$P_1 \triangleq$ present value of F_i

($1 \equiv$ present time period, aka Qtr 1)

$d \triangleq$ annual discount factor

$d' = \frac{d}{4} \triangleq$ quarterly discount factor

(Eqn (1) will be derived later [soon!])

Example : Qtr. 9

Profit or net cash flow } = \$1,750,000 = F_9
for Q9 (Year 3, Q1)

$$F_9 = \$1,750,000$$

Assume annual $d = 10\% \Rightarrow \underbrace{d'}_{\text{Qtr}} = \frac{d}{4} = 2.5\% \equiv 0.025$

$$\text{Present value of the } \left. \begin{array}{l} \text{future cash flow in Q9} \end{array} \right\} P_1 \stackrel{(1)}{=} \frac{F_9}{\left(1 + \frac{d}{4}\right)^{9-1}}$$

$$= \frac{\$1,750,000}{(1.025)^8}$$

$$= \$1,436,000 \equiv \$1,436K$$

Step 1(d) : Add up the present values :
(corresponding to each quarter) to
obtain the net present value
of expected PROFIT for
"x" years of the product-life
cycle, for an annual discount
factor of $d\%$.

$$\underline{\text{Net Present Value of Profit}} = \sum_{i=1}^n \text{Present Value}(Q_i)$$

$$\text{NPV} \stackrel{(1)}{=} \sum_{i=1}^n \frac{F_i}{\left(1 + \frac{d}{4}\right)^{i-1}}$$

For the text book example,

$$\left. \begin{array}{l} \text{NPV of the } \underline{\text{expected}} \\ \underline{\text{profit}} \text{ for 4 years} \\ \text{at a discount factor of } 10\% \end{array} \right\} \begin{array}{l} = \$8,203 \text{ k} \\ = \$8,203,000 \\ \underbrace{\hspace{1.5cm}} \\ \text{Base Case} \\ \text{or} \\ \text{Nominal} \end{array}$$

Step 2 : Perform a sensitivity analysis on the base-case (or nominal) model from Step 1, in order to quantitatively understand the effect of the input parameters (e.g., development cost, development time, ...) on expected profit

Step 3 : Use the sensitivity analysis performed on the base case model to understand & quantify the trade-offs between dev. time, dev. cost, production cost, & unit price

See the "Financial Modeling" Excel tutorial on Canvas;

and the case study in the text-book!

Ready to work on HW # 8, Prob. 3.